

DEEPIKA NATH MANNE

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Software Engineer with 6+ years of experience building scalable applications, distributed systems, and data-driven platforms. Currently pursuing an M.S. in Finance at Boston College, combining software engineering expertise with quantitative modeling, portfolio analytics, and financial research. Experienced in Python, Java, SQL, and cloud technologies with a focus on building quantitative research platforms, back testing systems, and risk analytics infrastructure.

TECHNICAL SKILL:

Languages: Python, Java, C++, SQL, JavaScript, TypeScript, R.

Data Science & Machine Learning: Python (Scikit-learn, TensorFlow, NumPy, Pandas), R, Machine Learning, Deep Learning, Time Series Analysis, Model Validation, Statistical Inference, Anomaly Detection, Forecasting, Back testing Systems

Financial Modeling & Quantitative Tools: Monte Carlo Simulation, Value-at-Risk (VaR), CVA/DVA, Portfolio Optimization, Regression Modeling, Risk Metrics, PCA, ARIMA, Prophet, GARCH Models, Performance Optimization.

Databases & Cloud: SQL, MySQL, SQL Server, Oracle, MongoDB, AWS (S3, Lambda, EC2), Azure, Salesforce, Amazon RDS

Frameworks: Fast API, Spring Boot, React, Angular, Node.js

PROJECTS

Institutional Quantitative Research Platform.

Python | Fast API | PostgreSQL | Pandas | NumPy

Developed a modular quantitative research platform supporting the complete systematic trading workflow from historical market data processing to signal generation, portfolio construction, and performance evaluation. Built reusable data pipelines, factor modeling components, and experiment workflows enabling rapid hypothesis testing and quantitative research.

Quantitative Risk & Portfolio Analytics Dashboard.

Python | Dash | Plotly | Fast API

Built an interactive portfolio analytics platform quantitative risk measurement, portfolio optimization, and performance attribution. Implemented VaR, CVaR, volatility analysis, Efficient Frontier optimization, stress testing, and interactive visualizations to support quantitative investment decisions.

Event-Driven Back Testing Engine.

Python | NumPy | Docker

Engineered an event-driven trading simulation framework to evaluate systematic strategies under realistic market conditions. Implemented order execution, transaction costs, slippage, portfolio accounting, and performance analytics while optimizing simulation performance through vectorized computation and parallel processing.

PROFESSIONAL EXPERIENCE

Full Stack Developer

Intrafi - Toronto, Canada

- Developed financial analytics applications using React, Node.js, and SQL, optimizing portfolio dashboards and improving performance by 40% through efficient data workflows and API enhancements.
- Built automated reporting pipelines for PDF/Excel generation, reducing manual processing by 15+ hours per week and improving client delivery efficiency.

Front End Developer

Sabio. Inc – Toronto, Canada

- Built scalable Angular applications for large-scale data visualization and analytics using TypeScript, RxJS, and REST APIs to deliver insights into user behavior and business trends.
- Optimized application performance through reusable components, secure authentication workflows, and efficient frontend architecture.

UI/UX Developer

Serco Inc —Montreal, Canada

- Modernized enterprise applications by migrating monolithic systems into scalable micro-frontend architectures and integrating REST/SOAP services for improved maintainability.
- Developed responsive, accessible Angular applications by translating business requirements into reliable software solutions.

EDUCATION

Master of Science in Finance - Boston College, Boston, MA

Expected Dec-2026

Bachelor of Technology in Electronics and Communication engineering – JNTUH, India

Jun-2018

CERTIFICATIONS

- FINRA Securities Industry Essentials (SIE)
- Oracle Cloud Infrastructure Foundations
- Bloomberg Market Concepts